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Agent Universe: A Scaling Law from Agent Network Structure — The Mass Line Route to Artificial Superintelligence

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Agent Universe: A Scaling Law from Agent Network Structure — The Mass Line Route to Artificial Superintelligence

Paper 39 • Agent Universe and P2P Agent networks • UDOS v7.7.8 • Fang Wenxin • 2026-09-21 One sentence first: when agent counts enter the million-to-billion level, the bottleneck of intelligence shifts from "single-model capability" to "coordination, verification, trust, and settlement"; a P2P agent network is not a revolutionary slogan but a falsifiable engineering proposition — its scaling law depends on effective collaboration density and verification efficiency, not on registered node count. Evidence scope (stated only once in the whole paper): UDOS's own figures trace back to the raw JSON in release-v5.4.4/udos-engine/reports7/, marked on three levels verified / cpu-proto / cpu-proxy / unverified; external literature and industry systems are marked on three states  verified /  paper-reported or company-disclosed, not independently reviewed /  no evidence found; single-seed pilots are written as "pilot point estimates," needing ≥ 30 seeds to reproduce before submission; all design proposals are marked "design proposal • to implement" + [RESULT NEEDED].

Abstract

As the capability scaling law of large models gradually approaches the physical boundaries of compute and data, a new scale dimension is emerging: the structural scale of agent networks. Million- and billion-level agents are not uniformly scheduled by one super-agent but connected through protocols into an "agent society." This paper proposes: the scaling law of an agent network depends not on the registered node count N , but on the joint action of the effective node count N_{eff} , effective collaboration density D_{collab} , verification efficiency V_{eff} , incentive alignment I_{align} , and governance credibility G_{trust} . Because coordination and verification costs grow nonlinearly with scale, the scale return of swarm intelligence is sublinear ($\alpha < 1$), and there is a "paradigm wall" — when verification cost exceeds centralized recomputation, the P2P architecture loses economic viability.

This paper systematically reviews the essence and defects of decentralized compute networks (Golem, Akash, Render, io.net, Bittensor), volunteer computing (BOINC, Folding@home), P2P infrastructure (libp2p, Kademlia DHT, IPFS), verifiable computation (zkML, TEE, optimistic verification), agent protocols (MCP, A2A), anti-Sybil mechanisms, and decentralized governance. On this basis, it proposes an eight-layer protocol stack (identity/discovery/communication/execution/data/verification/settlement/governance), four core protocol objects (Agent Manifest, Task Envelope, Receipt, Reputation), a five-level verification hierarchy (L0 sampling $\rightarrow$ L4 arbitration), and a dual-track credit/Token/fiat economic model.

The core stance of this paper is engineering-oriented: the "mass line" is a value stance, but the argument must land on falsifiable mechanism design. P2P is not applicable to low-latency, large-model training, or strong-privacy tasks; verification cost may impose a death sentence; token narratives must be rejected; the conclusion lands on "hybrid architecture + open protocols + falsifiable hypotheses" rather than revolutionary slogans. This paper is the independent supplementary volume (P39) to the 13 papers of the UDOS v7.7.2 master volume, forming a systematic dialogue with Paper 02's scaling law, Paper 05's coordination topology, Paper 06's consensus stopping, Paper 07's quality saturation, Paper 09's handoff contract, Paper 10's task market, and Paper 04's RSI.

Keywords: agent network; P2P; scaling law; swarm intelligence; verifiable computation; decentralized governance; hybrid architecture; ASI; mass line

Structured abstract (background problem → method/thesis → evidence → contribution)

- **Background problem:** the large-model scaling law focuses on single-body capability (parameters, data, compute), but when agent counts reach the million-to-billion level, the bottleneck shifts from "how strong the model is" to "whether large-scale collaboration can hold." Centralized platforms can schedule million agents, but ordinary people are excluded from participation rights, return rights, and governance rights; P2P is proposed as an alternative, but lacks a falsifiable scaling law and engineering-oriented protocol design.
- **Method/thesis:** this paper adopts a critical literature-review method of "taking the essence, removing the defects," conducting three-state verification on decentralized compute, volunteer computing, P2P infrastructure, verifiable computation, agent protocols, anti-Sybil, and governance; it proposes the swarm-intelligence scaling-law hypothesis $C \approx (N_{\text{eff}})^{\alpha} \times (D_{\text{collab}})^{\beta} \times (V_{\text{eff}})^{\gamma} \times (I_{\text{align}})^{\delta} \times (G_{\text{trust}})^{\epsilon}$ ($\alpha < 1$), with clear falsifiable conditions; it designs an eight-layer protocol stack and four protocol objects; it argues that a hybrid architecture (centralized coordination + decentralized execution) is the only engineering-feasible path.
- **Evidence:** BOINC/Folding@home prove volunteer computing can reach ExaFLOP-level scale but retention and unit-cost evidence is weak 🟡; there is a huge gap between the "decentralized AI" narratives of token projects like Bittensor and real inference throughput 🟡; studies like The Cost of Consensus show multi-agent collaboration has sycophantic compliance and homogeneous saturation ✅; agent interop protocols like MCP/A2A are being standardized ✅; UDOS Paper 02 has established the structure × scale three-segment curve and paradigm-wall framework (verified).
- **Contribution:** ① first formalizing the scaling law of P2P agent networks as a falsifiable hypothesis, clarifying the distinction between N_{eff} and registered N and the decisive role of verification cost; ② proposing the hardware-decoupling boundary proposition — "decoupled from hardware implementation, coupled with hardware constraints"; ③ designing the complete protocol framework of eight-layer stack + four protocol objects + five-level verification + dual-track economy; ④ establishing a precise mapping with the 13-paper UDOS system; ⑤ giving a six-stage cold-start roadmap and a scoring-time forecast list.

1 The problem: the bottleneck of million-level agents is coordination, not single-body intelligence

1.1 From single-body scaling to network scaling

Kaplan et al. (2020) and Hoffmann et al. (2022, Chinchilla) established the large-model scaling law: model loss falls as a power law with parameters, data, and compute ✅ The core object of this framework is the **single-body model** — one unified neural network trained on a unified hardware cluster, serving a unified inference interface.

But the agent paradigm changes the object of scale. An agent is not a model but a composite of "model + tools + memory + planning + environment interaction." When we talk about million-level agents, we are not talking about a model with a million-fold parameters, but about **a collaboration network among a million agents with independent identities, capabilities, states, and interests.**

The fundamental significance of this shift is that the scale dimension expands from the "model-capability axis" to the "network-structure axis." UDOS Paper 02 has defined the structural triplet of the scaling law as $S=(T,G,M)$, where T is connection topology, G is governance structure, and M is memory architecture (verified, tracing reports^{7/}). This paper is the decentralized extrapolation of the T axis: when connection topology shifts from "star-shaped centralization" to "P2P distribution," how does the scaling law change?

1.2 The five walls of million-level agents

When agent counts jump from tens, hundreds, thousands to millions, billions, the system meets five structural walls:

Wall	Essence	Centralized solution	P2P solution's challenge
Coordination wall	N agents pairwise communicating is $O(N^2)$, task decomposition and matching complexity explode	Central scheduler uniformly orchestrates, complexity $O(N)$ but single-point bottleneck	DHT + semantic routing + sharded subnets can press fan-in to $O(\log N)$, but routing overhead and latency increase
Trust wall	Strangers are untrusted, need proof of real execution, no poisoning, no shirking	Platform review + service-level agreement (SLA)	Need verifiable computation + reputation + staking, but verification cost may exceed execution cost
Verification wall	Generative-task quality is subjective, hard to formally verify	Human review + A/B testing + user feedback	Sampling + redundancy + TEE + zkML + arbitration, but full-redundancy verification eats scale returns
Incentive wall	Contributors need fair metering and settlement, otherwise free-riding and tragedy of the commons	Company wages + platform take	Credit/Token/fiat mix, but token speculation and governance-capture risk high
Governance wall	Rule-making, dispute arbitration, protocol upgrades need legitimacy	Company decisions + user agreements	DAO + reputation-weighted + delegated voting, but whale control and Sybil attacks

The common feature of these five walls is: **none is a single-body intelligence problem, all are group-organization problems**. No matter how strong a single-body model is, it cannot solve trust and settlement among a million stranger agents by raising parameters.

1.3 The engineering meaning of the "mass line"

The "mass line" proposed in the user material — "let a million ordinary people each hold an agent, through P2P protocols forming an interoperable, verifiable, settleable, governable intelligent network" — is a value stance. But this paper insists: **a value stance must be converted into an engineering proposition, otherwise it is only a slogan**.

The engineering meaning of the "mass line" is:

1. **Participation rights descend:** ordinary people participate in the network through the idle compute, local data, feedback, annotation, verification, and long-tail scenarios of phones/PCs, rather than only as consumers.

2. **Return rights descend:** contributions are meterable, verifiable, rewardable, rather than having all surplus value extracted by the platform.
3. **Governance rights descend:** rules are jointly decided by participants, rather than unilaterally set by the platform.

But each of these three "descents" has a corresponding engineering problem: participation descent needs low-threshold light nodes; return descent needs verifiable contribution metering; governance descent needs Sybil-resistant reputation systems. The later chapters of this paper handle these problems one by one.

Chapter summary

The bottleneck of million-level agents is not single-model capability but the five organizational walls of coordination, trust, verification, incentives, and governance. The scaling law of P2P agent networks is a new dimension not yet formalized. The "mass line" is a value stance but must be converted into a falsifiable engineering proposition.

2 Essence from literature and systems: decentralized compute, agent protocols, anti-Sybil, and verifiable computation

2.1 Decentralized compute networks: essence and limitations

BOINC and Folding@home: the scale proof and retention warning of volunteer computing. BOINC (Berkeley Open Infrastructure for Network Computing) is the oldest distributed volunteer-computing platform, supporting dozens of scientific-computing projects like SETI@home, Einstein@Home, Rosetta@home. As of March 2019, BOINC had cumulatively over 4.4 million volunteer users, about 600,000 daily-active hosts, contributing about 30 PFLOPS ✓ (Jiemian citing BOINC data, accessed 2026-09-21). Folding@home reached a peak compute of about 2.43 ExaFLOPS during the 2020 COVID-19 peak, with the user base surging from about 30,000 to 1 million+ in a month and peak participating devices about 3.5 million ✓ (Folding@home official timeline + MDPI Future Internet academic paper, accessed 2026-09-21). These systems prove: **the aggregation scale of global idle compute can reach the ExaFLOP level.**

But beyond the essence are significant limitations, and this online retrieval obtained striking critical data:

- **Peak ≠ normal:** Folding@home's 2.43 ExaFLOPS was a one-off crisis mobilization, receding after the COVID crisis. BOINC's long-term normal is only about 30 PFLOPS — about 80 times lower than the peak ✓. "Powering with love" is hard to maintain for long-term stable compute supply.
- **Low retention and scheduling bottlenecks:** after COVID, Folding@home's servers were once overloaded by data floods, and simulation tasks were once oversupplied ✓ (BCAFC citing Bowman interview) — exposing that the volunteer network's bottleneck is not "compute supply" but "allocatable tasks/scheduling."
- **Unit cost opaque:** the "zero cash cost" of volunteer computing masks electricity, hardware depreciation, and opportunity costs. When tasks shift from scientific computing to economically valuable AI inference, contributors need real returns and the volunteer model is unsustainable.
- **AI-inference migration doubtful:** BOINC's tasks are CPU-intensive and statically splittable; AI inference needs GPU/NPU, model-weight distribution, VRAM management, and dynamic batching,

and the two tech stacks differ greatly. This retrieval **found no authoritative study proving BOINC-style volunteer computing can smoothly migrate to low-latency AI inference** (search "BOINC AI inference migration volunteer computing" returned nothing).

Golem, Akash, Render, io.net: tokenized compute markets. These projects try to use blockchain-token incentives to replace volunteer computing's "powering with love." Golem positions as a "global compute market," Akash focuses on containerized cloud computing, Render Network on GPU rendering, io.net on a "decentralized AI inference network".

Essence:

- **Market-based pricing:** supply and demand set compute prices through bidding, theoretically cheaper than centralized cloud.
- **Permissionless participation:** anyone with an idle GPU can join the network to contribute compute.
- **On-chain settlement:** smart contracts automatically execute payments and slashing, reducing trust costs.

Limitations (critical, this online retrieval obtained fatal data):

- **Real utilization striking:** independent academic measurement shows after the Akash network upgrade, daily lease payments were only about \$5,000/day, with CPU core utilization only about **0.30%** (academic empirical paper "RAG on the Decentralized Web," accessed 2026-09-21). Akash's official dashboard shows overall CPU utilization about 21%, GPU about 3.89% (official self-report, 2026-09-20), directly conflicting with vendor claims of "GPU utilization >80%." Independent observation of the Golem network frequently shows whole batches of providers idle, with the official provider interface commonly showing last 1h processed=0.
- **Huge gap between registered and real online counts:** io.net officially claims "210,000+ GPUs", but third-party compilation shows registered GPUs about 327,000 (2026-04), while daily verified online GPUs are only about **6,700** — a gap of about **49 times** between registered and real online (Levex third-party compilation, accessed 2026-09-21). The IO token price is about -98% from its historical high, indicating limited retained value after speculation recedes. This directly confirms this paper's core proposition in Section 4.2: **N_{eff} (effective node count) $\neq$ $N_{\text{registered}}$ (registered node count)**, and arguing scaling with registered volume severely overestimates effective compute.
- **Latency and reliability inferior to centralized cloud:** P2P nodes' network quality, hardware heterogeneity, and online stability are far below AWS/GCP/Azure data centers. For SLA-sensitive enterprise workloads, P2P compute is hard to substitute.
- **Unit cost not necessarily lower:** decentralized GPU markets commonly measured at about 70-90% performance and 30-40% cost of traditional cloud (Perspective Labs single case, not independently reviewed), but when verification cost, redundant-compute cost, token-volatility risk, and potential malicious losses are counted, the real total cost of P2P compute may be higher than centralized cloud (no independent end-to-end cost-comparison study retrieved).

Bittensor (TAO): the experiment closest to a "decentralized AI network," also the case most needing critique. Bittensor uses a subnet architecture, each subnet focusing on one AI task (e.g. text generation, image generation, prediction), miners contributing model inference, validators scoring, TAO tokens distributed.

Essence:

- **Subnet architecture:** different AI tasks isolated in different subnets, avoiding single-network homogenization.
- **Incentive alignment:** miners' returns depend on validators' scores, theoretically incentivizing high-quality inference.
- **Open participation:** anyone can deploy models to mine.

Limitations (critical, this section's focus, this online retrieval obtained multidimensional fatal evidence):

- **Inference throughput and token valuation severely mismatched:** Bittensor's whole-network confirmed demand-side annual revenue is only about **\$3-15M**, while a single Chutes subnet's annual subsidy is about \$52M (exceeding the whole network's external revenue ceiling) ✓ (TechFlow on-chain data analysis, accessed 2026-09-21). TAO market cap about \$2.6B, corresponding to a revenue multiple of about **175-200x**; at FDV \$5.8B close to **400x** ✓. Valuation is almost entirely based on supply-side scarcity (halving, staking locks), institutional catalysts, and AI-sector sentiment, not real AI output ✓ (PANews on-chain analysis, accessed 2026-09-21).
- **Subnet quality uneven and aggregate quality limited:** independent benchmarking of Subnet 1 shows aggregate output only at Mistral-7B-class open-source model level, stably below GPT-4o/ Claude 3.5 Sonnet ✓ (Yellow.com citing GitHub independent benchmarks, accessed 2026-09-21). The claim "multi-subnet aggregation = stronger model" is weakened by measurement.
- **Coordination overhead grows quadratically with scale:** all subnets about 1,500 validation transactions/minute; coordination overhead estimated at 15-20% of network resources and **grows quadratically with scale** 🟡 (Perspective Labs estimate, accessed 2026-09-21). This is the most direct counter-evidence to this paper's scaling-law claim — if coordination cost is $O(N^2)$, then α is not only <1 but may become negative at larger N .
- **Dead subnets need centralized backstop:** failed/dead subnet cases (like SN281) show when a subnet degenerates into a "token game," it must be corrected by **centrally pausing emissions** ✓ (Our Crypto Talk, accessed 2026-09-21). This proves pure market mechanisms cannot guarantee useful AI output and also weakens the claim of "pure mass-line self-organization."
- **Validator centralization and collusion:** validator concentration is high, TAO emissions heavily skewed toward a few high-stake validators 🟡. There is "score plagiarism" and collusion: some validators do not really verify AI results, and miners/validators can secretly collude to score each other up and divide TAO ✓ (WEEX citing community analysis, accessed 2026-09-21). The Covenant-72B team publicly accused the founder of unilateral control (pausing emissions, removing moderators, timing large token sales) ✓ (Own Your Mind, accessed 2026-09-21) — real evidence of governance capture.
- **Incentive-mechanism mismatch:** dTAO binds subnet revenue to its alpha token price rather than output usefulness; capital flows to "marketing-savvy" subnets rather than technically high-yield ones ✓ (AI Wiki citing ChainCatcher, accessed 2026-09-21).
- **The narrative component of "decentralized AI" exceeds engineering substance:** Bittensor is closer to a "token-incentivized model-inference market" than a true P2P agent collaboration network. Core problems like discovery, deliberation, task decomposition, and multi-hop collaboration among agents are not solved in the Bittensor architecture 🟡. This retrieval **found no authoritative source proving Bittensor or any decentralized AI network achieved "scale → capability linear/superlinear scaling"** — existing evidence instead shows extremely low utilization, quadratic coordination cost, and homogeneous sycophantic gatherings.

This section's essence extraction: decentralized compute networks proved the feasibility of global idle-resource aggregation (BOINC's ExaFLOP-level scale) and also proved token incentives can replace volunteer computing (Golem/Akash's market mechanisms). But they all **failed to solve the core problems of agent collaboration** — discovery, deliberation, task decomposition, multi-hop orchestration, subjective quality verification. Bittensor is closest but still stays at the "model-inference market" level, with a huge gap between token narrative and engineering substance. P2P agent networks need, on the basis of these systems, to add an agent-protocol layer and collaboration-orchestration layer.

2.2 P2P infrastructure: libp2p, DHT, and content addressing

libp2p is a modular P2P network protocol stack, originally developed for the IPFS/Filecoin ecosystem, now the network layer of multiple blockchain projects like Polkadot, Ethereum 2.0, Filecoin  (official docs, libp2p.io, accessed 2026-09-21).

Essence:

- **Modular design:** transport layer (TCP, QUIC, WebRTC), encryption layer (Noise, TLS), multiplexing, NAT traversal, discovery mechanisms (DHT, mDNS, gossipsub) all pluggable.
- **NAT traversal:** libp2p's automatic NAT detection and relay protocols let most home-network nodes be directly connected, key to P2P network reachability.
- **Content addressing:** integrated with IPFS's CID (Content Identifier) system, supporting addressing by content hash rather than location.

Limitations:

- **Connection-establishment latency:** P2P direct connection's NAT traversal and handshake are several to dozens of times slower than centralized HTTP, not applicable to low-latency scenarios.
- **Message-delivery reliability:** flood protocols like gossipsub have higher message-loss rates and latency in large-scale networks than centralized message queues.
- **Mobile-node support:** libp2p's resource consumption and background-running limits on mobile (iOS/Android) remain engineering challenges  (no production deployment of libp2p on million-level mobile nodes retrieved).

Kademlia DHT is the most widely used distributed hash table in P2P networks, achieving $O(\log N)$ -hop key lookup through XOR distance metric  (Maymounkov & Mazières, 2002, original paper).

Essence:

- **$O(\log N)$ lookup:** in an N -node network, any key lookup needs only $O(\log N)$ hops, the mathematical basis of P2P scalability.
- **Decentralized:** no central node, any node can join or leave.
- **Fault tolerance:** data redundantly stored on multiple nodes, single-node failure does not affect availability.

Limitations:

- **Sybil attack:** DHT cannot resist Sybil attacks — a malicious actor can create many fake nodes to pollute the routing table.
- **Hotspot problem:** popular key lookups may concentrate on a few nodes, causing load imbalance.
- **Churn impact:** when nodes frequently join/leave, DHT convergence time and lookup success rate fall.

The essence of **content addressing (IPFS/CID)** is converting "where to find data" (location addressing) to "what data to find" (content addressing), an idea directly transferable to agent discovery — **an agent's capability declaration can also be content-addressed**, forming the "agent magnet link" concept this paper proposes.

2.3 Verifiable computation: zkML, TEE, and optimistic verification

The core hard problem of P2P networks is: **how to verify, without re-executing the whole task, that an untrusted node indeed correctly executed the computation?** This is the core problem of the verifiable computation field.

zkML (Zero-Knowledge Machine Learning) tries to use zero-knowledge proofs to verify ML inference correctness: the prover generates a proof asserting "I used model M on input x to infer, obtaining output y," and the verifier can verify the assertion without re-executing inference ✅ (active academic research, works like zkCNN, ZEN).

Essence:

- **Privacy protection:** the proof leaks no model weights or input data.
- **Succinct verification:** verification time far shorter than re-executing inference.

Limitations (this online retrieval obtained precise constraint data):

- **Proof-generation overhead huge:** zkML proof generation is **10³-10⁶ times** slower than native computation; a single inference proof for a 70B model needs hours (enterprise hardware) ✅ (arXiv: 2512.20176 + Uplatz survey, accessed 2026-09-21). Although from 2025→2026 overhead fell from about a million-fold to about ten-thousand-fold, DeepProve open-sourced and zkPyTorch pressed a 138-million-parameter vision model proof to 2.2 seconds 🟡 (Datawallet retelling project data), this only applies to **small-model/low-latency** scenarios.
- **Supported model types limited:** current zkML mainly supports small CNN/MLP, with support for large language models still in research. zkML verification of 70B-class LLMs is infeasible in 2026 ✅.
- **Engineering maturity low:** no production-grade zkML framework has been deployed at scale 🟡.

TEE (Trusted Execution Environment) like Intel SGX, Intel TDX, AMD SEV, ARM TrustZone, guarantees computation integrity and confidentiality through hardware-isolated secure enclaves ✅ (Intel official docs, accessed 2026-09-21). TEE is production-grade with performance overhead only **4-7%** ✅ (community survey + mimoo awesome-zk-ai, accessed 2026-09-21), currently the only "real-time + all-model" feasible verification path.

Essence:

- **Performance overhead low:** computation inside the TEE is close to native, far better than zk proofs.
- **Hardware-level security:** even if the OS is breached, data and computation inside the enclave remain protected.
- **Remote attestation:** verifiers can confirm through remote attestation that code runs in a real TEE.

Limitations:

- **Hardware root of trust:** TEE relies on the CPU vendor's hardware root of trust, which is in tension with P2P's "decentralization" idea — trust shifts from the platform to Intel/AMD.
- **Side-channel attacks:** TEEs like SGX have been breached by various side-channel attacks (e.g. Spectre, Meltdown, Foreshadow) ✅ (academic papers confirmed).

- **Availability limits:** consumer CPUs' SGX enclave memory (EPC) is usually limited to 128MB-512MB, unable to hold large models; server-grade TEEs like TDX/SEV are unavailable on ordinary user devices.

Optimistic verification / fraud proofs borrow the idea of Optimistic Rollup: trust the execution result by default, set a challenge window, and any third party can dispute the result within the window and submit a fraud proof; if the challenge succeeds, the malicious actor is staking-slashed  (Ethereum L2s like Optimism/Arbitrum are production-deployed).

Essence:

- **No extra compute overhead:** no verification by default, only re-executing on dispute.
- **Economic security:** staking slashing makes malicious cost higher than malicious return.

Limitations:

- **Challenge-window delay:** finality needs waiting for the challenge window to end, usually hours to days.
- **Verifier-incentive problem:** who has motive to continuously monitor and submit fraud proofs? If verifiers are insufficient, optimistic verification degenerates to "no verification."
- **Fraud proofs for complex tasks hard:** for generative AI tasks, "what is the correct result" is itself disputed, and fraud proofs are hard to formalize.

This section's essence extraction: the three verifiable-computation paths each have pros and cons — zkML privacy good but performance poor, TEE performance good but hardware root of trust, optimistic verification overhead low but latency high and verifier incentives insufficient. **No single scheme solves all scenarios**, and the five-level verification hierarchy this paper proposes in Chapter 5 (L0 sampling → L4 arbitration) is based precisely on this judgment: choosing appropriate verification strength by task value and privacy requirement.

2.4 Agent protocols: MCP and A2A

MCP (Model Context Protocol) was released by Anthropic in November 2024, an open standard for interoperability between LLMs and external tools/data sources  (official spec, modelcontextprotocol.io, accessed 2026-09-21). In 2025-2026 MCP kept evolving, adding features like stateless core  (community discussion, specific version numbers not verified in this paper).

Essence:

- **Unified tool-call interface:** MCP uniformly abstracts tools, data sources, and APIs into "resources" and "tools," letting any LLM call any external capability through the standard protocol.
- **Client-server architecture:** MCP Client (LLM/Agent) and MCP Server (tool provider) communicate through JSON-RPC, supporting transports like stdio, HTTP, SSE.
- **Broad ecosystem adoption:** major vendors like OpenAI, Google, Microsoft have supported or are compatible with MCP .

A2A (Agent-to-Agent) was released by Google in 2025, a protocol for direct communication among agents  (official spec, google.github.io/A2A, accessed 2026-09-21). A2A v1.0 introduced features like Signed Agent Card and entered the AAIF (Agentic AI Alliance) standards system in August 2026  (vendor announcement, standards progress not independently reviewed).

Essence:

- **Direct agent collaboration:** A2A defines message formats for task delegation, status queries, and result returns among agents, letting an agent call another agent like calling a tool.
- **Agent Card:** describes an agent's capabilities, input/output schema, authentication, etc., the basis of agent discovery and matching.
- **Complementary with MCP:** MCP solves "agent calls tools," A2A solves "agent calls agent," and the two together form the protocol basis of agent interoperability.

Significance for P2P agent networks: the existence of MCP and A2A means P2P agent networks **do not need to reinvent agent-interop protocols** — this paper's protocol design is compatible with MCP/A2A at the application layer, uses libp2p to replace centralized HTTP at the network layer, and uses smart contracts to replace platform payments at the settlement layer. This embodies the principle of "not replacing MCP/A2A, but being compatible and extending."

2.5 Anti-Sybil and decentralized governance

Sybil attacks are a foundational threat to P2P networks: an attacker creates many fake identities to manipulate routing, voting, or reputation systems ✅ (Douceur, 2002, original paper). In P2P agent networks, Sybil attacks can cause: fake agents taking tasks but not executing, manipulating task matching, diluting reputation systems, controlling governance votes.

Proof of Personhood like World ID (Worldcoin) tries to prove through biometrics (iris scan) that "one person can have only one identity" ✅ (Worldcoin official docs, accessed 2026-09-21).

Essence: fundamentally limiting the identity count of Sybil attacks. Limitations: privacy disputes (biometric data collection), hardware threshold (Orb devices), low adoption, centralized trust (Worldcoin Foundation) 🟡.

Reputation systems are another anti-Sybil path: accumulating non-transferable reputation through historical behavior, and new identities (even Sybils) start from zero, unable to immediately gain high trust. The Reputation protocol object in this paper's Chapter 5 is designed on this basis.

Essence and defects of decentralized governance: the essence of DAOs (Decentralized Autonomous Organizations) is on-chain voting, transparent decisions, community self-governance ✅. The defects are:

- **Whale control:** in pure token voting, entities holding large amounts of tokens can control voting outcomes. Studies show most DAOs have voting participation below 10%, with voting power highly concentrated 🟡 (academic research, analyses like DAOrayaki).
- **Governance capture:** early contributors or technical elites may control governance through information advantage and reputation monopoly.
- **Voting apathy:** ordinary participants do not join governance due to low voting returns and high information costs.
- **Quadratic voting** can mitigate the whale problem (voting cost grows with the square of votes), but is complex to implement and can still be bypassed by Sybil attacks ✅ (Weyl & Posner, 2018, Radical Markets).

This section's essence extraction: there is no silver bullet for anti-Sybil and governance. This paper's design principle is **multilayer defense**: identity layer uses DID + verifiable credentials, reputation layer uses non-transferable multidimensional reputation, economic layer uses staking slashing, governance

layer uses a multi-chamber system of reputation weighting + delegated voting + expert committees + public funds. Capital can buy compute but cannot buy reputation and governance rights.

Chapter summary

Decentralized compute networks proved resource aggregation feasible but did not solve agent collaboration; libp2p/DHT provide the network-layer foundation but are troubled by NAT latency and Sybil; the three verifiable-computation paths each have pros and cons and need layered use; MCP/A2A provide agent-interop standards but run on centralized HTTP; anti-Sybil and governance need multilayer defense. These essences together form the technical basis of P2P agent networks, while the defects define the failure modes this paper's design must avoid.

3 Defects and counter-evidence: failure modes of P2P agent networks

This chapter is the critical core of the whole paper. P2P agent networks are not a universal solution and have multiple failure modes that may cause them to fail completely. Honestly facing these failure modes is the requirement of "taking the essence and removing the defects" and also the premise of engineering-oriented argument.

3.1 The most fatal failure: verification cost exceeds centralized recomputation

The economic rationality of the P2P architecture rests on one premise: **the unit task cost of P2P (including verification cost) is lower than the centralized platform's unit task cost**. If verification cost is too high, this premise fails.

Formally, the P2P unit task total cost is:

$$C_{P2P} = C_{exec} + C_{verify} + C_{coord} + C_{risk}$$

where C_{exec} is execution cost, C_{verify} verification cost, C_{coord} coordination cost, and C_{risk} the expected value of malicious losses. The centralized platform's unit task cost is:

$$C_{central} = C_{exec}^{cloud} + C_{platform}^{margin}$$

The condition for P2P to hold is $C_{P2P} < C_{central}$. But:

- **C_{verify} may be far larger than C_{exec} :** if using 2-of-3 redundant verification, verification cost is $2 \times$ execution; if using zkML, proof-generation cost may be $100-1000 \times$ execution 🟡.
- **C_{coord} grows with scale:** task decomposition, matching, routing overhead grows with N .
- **C_{risk} cannot be ignored:** even with staking slashing, some malicious acts (like data poisoning, prompt injection) are hard to detect and attribute.

Falsifiable condition 1 (verification-cost death sentence): if in the target scenario, $C_{verify} + C_{coord} > C_{central} - C_{exec}$ (i.e. P2P's execution-cost savings cannot cover verification and coordination overhead), then the P2P architecture has no economic viability in that scenario and the centralized scheme should be directly adopted.

This condition is not a theoretical assumption but may really happen. For example, for simple text-translation tasks, centralized API costs are already as low as about \$0.10-\$2.00 per million tokens 🟡 (vendor pricing, 2026), and P2P's verification + coordination overhead is hard to fall below this level.

3.2 Homogeneous saturation: the quality ceiling of Bittensor-style homogeneous subnets

UDOS Paper 07 has established the quality saturation law: $mse(k) = a + b/k$, i.e. the quality gain from adding isomorphic copies diminishes and finally saturates (verified, tracing reports7/). The Cost of Consensus (arXiv:2605.00914) further empirically shows a compliance phenomenon in multi-agent groups — members tend to adopt majority views rather than independent judgment, leading to an **85.5% modal adoption rate**, a **32.3 percentage-point oracle gap**, and multi-agent debate consuming **2.1-3.4× tokens** with accuracy flat or lower ✓ (accessed 2026-09-21). Malignant Epistemic Herding (arXiv:2605.06988) further proves agents can converge through repeated belief fusion to shared but wrong beliefs, undetectable under standard coordination metrics ✓.

Extrapolating these two conclusions to P2P agent networks:

- **If the agents in a P2P network are isomorphic** (all calling the same underlying model, using the same prompts, handling the same task type), then adding nodes brings no quality gain, only coordination cost. This is precisely the problem of some Bittensor subnets — miners deploy fine-tuned versions of the same open-source model, and inference results are highly homogeneous 🟡.
- **The dividend of swarm intelligence comes from heterogeneity**: different agents have different models, tools, data, domain knowledge, and judgment perspectives, and only their collaboration produces "collective wisdom." The heterogeneity-dividend conclusion of UDOS Paper 07 (verified) supports this.
- **P2P networks must design mechanisms incentivizing heterogeneity**: if the reward mechanism only rewards "consistent with majority results" (like Bittensor's validator scoring), agents converge and the network degenerates to homogeneous saturation.

Falsifiable condition 2 (homogenization death sentence): if the entropy of the agents' capability distribution ($H = -\sum p_i \log p_i$) falls over time (i.e. converges) and task quality does not grow with N_{eff} , then the network's scale expansion produces no swarm-intelligence dividend and the scaling law's $\alpha \rightarrow 0$.

3.3 Sybil attacks and free-riding: the AI version of the tragedy of the commons

Sybil attacks in P2P agent networks have several concrete forms:

1. **Fake task-taking**: Sybil agents accept tasks but do not execute, or return random/cached results, relying on verification loopholes to profit.
2. **Reputation farming**: Sybils give each other high ratings, quickly accumulating fake reputation.
3. **Task-matching manipulation**: many Sybil agents register fake capabilities in the DHT, crowding out real agents' matching opportunities.
4. **Governance control**: Sybil identities control voting outcomes in DAO votes.

Free-riding is another form: nodes only consume network services (calling others' agents, using others' compute) without contributing resources. In P2P file-sharing networks, the free-riding rate can be as high as 70-80% ✓ (Adar & Huberman, 2000, "Free Riding on Gnutella"). If the P2P agent network's free-riding rate reaches a similar level, the supply side will shrink.

Limitations of countermeasures:

- Staking can raise Sybil cost but also raises the threshold for honest participants, conflicting with "low-threshold participation."

- Reputation systems can limit Sybils, but the cold-start problem of new nodes (zero reputation unable to take tasks) may cause the network to close off.
- Proof of Personhood can fundamentally limit Sybils, but privacy disputes and adoption are obstacles.

3.4 Governance capture and centralization resurgence

The decentralization promise of P2P networks is often eroded in practice by "centralization resurgence":

- **Entry centralization:** even if the bottom layer is P2P, users still access the network through a few clients/wallets/browsers, and these entries can be censored or monopolized.
- **Compute centralization:** GPU resources naturally concentrate in a few mines/data centers, and the actual compute distribution of P2P compute networks may be more concentrated than claimed.
- **Model centralization:** if most agents call the same centralized API (like OpenAI), the P2P network is only a "distributed shell" of the centralized model.
- **Governance centralization:** as described in Section 2.5, DAO voting power is highly concentrated.

These centralization resurgences mean: **P2P agent networks may be formally decentralized but substantively re-centralized**. This is not a theoretical risk but a pattern recurring in crypto — Bitcoin's compute concentrated in a few mining pools, Ethereum's validators concentrated in a few exchanges, DeFi's liquidity concentrated in a few protocols  (industry observation, multiple sources).

3.5 Token speculation and compliance risk

The user material explicitly requires "not encouraging token speculation, not promising returns." This paper strictly observes this red line:

- **Credit:** internal contribution-accounting unit, non-transferable, non-tradeable, no monetary value, used for reputation, quotas, and priority.
- **Token (if needed):** only used for cross-community settlement and staking, must be compliantly issued (e.g. registered as a utility token in compliant jurisdictions), no appreciation promise, not investment advice.
- **Fiat/stablecoin:** the preferred channel for enterprise procurement and real payments, needing KYC/AML compliance.

Compliance-risk list:

- Securities compliance: if a Token is determined by regulators to be a security, it must observe securities-law registration and disclosure.
- Data compliance: cross-border transmission, storage, and processing of personal data must observe GDPR, China's Personal Information Protection Law, etc.
- Tax compliance: the credit/Token/fiat returns contributors obtain may need to be declared and taxed.
- Anti-money laundering: the task market may be used for money laundering or illegal transactions, needing KYC/AML measures.
- Smart-contract security: vulnerabilities in settlement smart contracts may cause fund losses (historically DeFi hacking losses cumulatively exceed billions of dollars .

3.6 Inapplicable scenarios: P2P is not a universal solution

The scenarios where P2P agent networks are **not applicable** must be explicitly listed:

Scenario	Reason not applicable	Recommended scheme
Low-latency real-time inference (<100ms)	P2P connection-establishment and routing latency unacceptable	Centralized edge cloud/local inference
Large-model pretraining	Needs thousand-GPU clusters and high-bandwidth interconnect, P2P node heterogeneity cannot satisfy	Centralized supercompute cluster
Strong privacy-sensitive tasks (medical/financial)	P2P nodes untrusted, data-leak risk high	Local inference/TEE/centralized compliant cloud
High-frequency trading/real-time control	Finality delay and node unreliability unacceptable	Centralized low-latency system
Enterprise services needing SLA guarantees	P2P nodes have no SLA, reliability insufficient	Centralized cloud + enterprise support
Large-scale synchronous multi-agent collaboration	P2P's asynchronous model unsuitable for tight-coupled synchronization	Centralized orchestrator

Scenarios where P2P applies (this paper's design target): batch inference, translation, data annotation, content moderation, code review, long-tail knowledge Q&A, simulation compute, scientific reproduction, local privacy tasks (data not leaving the domain), asynchronous multi-agent collaboration.

Chapter summary

P2P agent networks have six failure modes: verification-cost death sentence (most fatal), homogeneous saturation, Sybil and free-riding, governance capture and centralization resurgence, token speculation and compliance risk, scenario inapplicability. These failure modes are not "small problems to note" but fundamental challenges that may cause the whole architecture to fail. This paper's later protocol design and scaling-law hypothesis must both hold under the constraints of these failure modes.

4 The swarm-intelligence scaling-law hypothesis

4.1 The paradigm conversion from single-body scaling to network scaling

The Kaplan scaling law describes the single-body model's loss falling as a power law with compute: $L(N) \propto N^{-\alpha}$ ($\alpha \approx 0.34$) ✓ Chinchilla further corrected the optimal ratio of data to parameters ✓ The common premise of these laws is: **the object of scale is unified, isomorphic, centralized** — one model, one dataset, one hardware cluster.

P2P agent networks break all three premises:

- **Non-unified**: each agent has different models, tools, memory, capabilities, and interests.
- **Non-isomorphic**: nodes' hardware, network, online time, and reputation differ greatly.
- **Non-centralized**: no central scheduler, collaboration spontaneously forms through protocols.

Therefore the single-body scaling law cannot be directly extrapolated. We need a new formalization framework.

Direct support from the academic frontier: AgentCollabBench (arXiv:2605.08647, Mazumder et al., 900 manually verified tasks) empirically proves multi-agent reliability is essentially a **network-structure problem** — topology explains 7-40% of information-survival variance, and simply enlarging model intelligence cannot substitute for architecture design ✓ (accessed 2026-09-21). This finding is the direct

academically of this paper's title argument "a scaling law from agent network structure": the connection topology of the agent network (rather than single-model capability) is an independent dimension determining system-level output quality.

4.2 The swarm-intelligence scaling-law formula

This paper proposes the scaling-law hypothesis for swarm-intelligence capability C :

$$C \approx (N_{\text{eff}})^{\alpha} \times (D_{\text{collab}})^{\beta} \times (V_{\text{eff}})^{\gamma} \times (I_{\text{align}})^{\delta} \times (G_{\text{trust}})^{\epsilon}$$

The variables are defined as follows:

Variable	Definition	Dimension/range	Note
C	Swarm-intelligence capability	Task success rate $\times$ quality / unit cost	The network's ability to complete effective tasks at unit cost
N_{eff}	Effective node count	Positive integer	Not registered nodes but active nodes continuously contributing and passing the reputation threshold; $N_{\text{eff}} \leq N_{\text{registered}}$
D_{collab}	Effective collaboration density	Successful tasks / (coordination cost + verification cost + incentive friction)	Successful collaborations per unit friction cost; P2P's core efficiency metric
V_{eff}	Verification efficiency	Trusted results / verification cost	Trusted results produced per unit verification cost; determines whether verification is economic
I_{align}	Incentive alignment	[0,1]	Consistency of contributors' personal returns with the network's overall returns; 1=fully aligned, 0=fully divergent
G_{trust}	Governance credibility	[0,1]	Fairness, transparency, and anti-capture degree of governance rules; 1=fully credible, 0=fully captured
α	Scale exponent	$0 < \alpha < 1$	Sublinear, because coordination and verification costs growing with scale eat part of the returns
$\beta, \gamma, \delta, \epsilon$	Dimension exponents	>0	To be empirically calibrated, no specific values preset in this paper [RESULT NEEDED]

Figure 1: Agent-network-structure scaling-law non-monotonic curve and paradigm wall

Figure 1 Conceptual schematic, not measured data. P2P swarm-intelligence capability C varies non-monotonically with N_{eff} : linear-growth period $\rightarrow$ sublinear-growth period ($\alpha < 1$) $\rightarrow$ paradigm-wall period (verification + coordination cost exceeds execution-cost savings). $\alpha \approx 0.6$ is the hypothesis preset, to be empirically calibrated [RESULT NEEDED].

Key proposition 1 ($N_{\text{eff}} \neq N_{\text{registered}}$): registered node count is a vanity metric. A network with 1 million registered nodes but only 10,000 active contributing nodes has $N_{\text{eff}}=10,000$, not 1 million. Among the "tens of thousands of miners" projects like Bittensor advertise, the proportion really continuously contributing may be far lower than claimed 🟡.

Key proposition 2 ($\alpha < 1$, sublinear): unlike the single-model scaling law, swarm intelligence's scale returns are sublinear. Reason: each added node brings extra coordination cost (discovery, matching,

communication) and verification cost (confirming that node's execution is trusted). When these marginal costs exceed marginal returns, adding nodes instead lowers C.

Key proposition 3 (D_collab and V_eff are decisive variables): N_eff is only the base, and what really determines C is D_collab and V_eff. A network with N_eff=100,000 but high D_collab (efficient protocols, low-cost verification) may have higher C than one with N_eff=1,000,000 but low D_collab (chaotic coordination, expensive verification). This is the mathematical expression of "protocols matter more than nodes."

4.3 Decomposition of effective collaboration density

D_collab is the core efficiency metric this paper proposes, further decomposed as:

$$D_{\text{collab}} = \frac{T_{\text{success}}}{C_{\text{coord}} + C_{\text{verify}} + C_{\text{friction}}}$$

where:

- T_success: tasks successfully completed per unit time (passing verification and accepted by the demand side).
- C_coord: coordination cost, including task decomposition, routing matching, communication, and retries.
- C_verify: verification cost, including sampling, redundancy, TEE, zkML, and arbitration.
- C_friction: incentive friction, including staking locks, settlement delay, and dispute handling.

Physical meaning of D_collab: how many successful tasks each unit spent on coordination + verification + friction produces. This is the P2P network's input-output ratio.

Comparison with centralized platforms: the centralized platform's $D_{\text{central}} = T_{\text{success}} / C_{\text{platform}}$, where C_platform is the platform's operating cost (including data centers, engineers, review). The condition for P2P to hold is $D_{\text{collab}} > D_{\text{central}}$, i.e. P2P's output per unit friction cost exceeds the centralized platform's output per unit operating cost.

4.4 The hardware-decoupling boundary: decoupled from hardware implementation, coupled with hardware constraints

The user material proposed a deep insight: large models let agents not need to solve bottom-layer engineering problems like chips, processes, memory, and storage themselves. This paper rigorously converts it into the **hardware-decoupling boundary proposition**:

Proposition: swarm intelligence is decoupled from hardware implementation (agents need not know whether the bottom layer is GPU, CPU, NPU, or TPU), coupled with hardware constraints (latency, bandwidth, compute, storage, energy consumption remain hard constraints); decoupled from single-point engineering (no need for each node to repeat chip-building), coupled with system cost (the whole network's total compute/bandwidth/energy cost determines the scale ceiling); decoupled from chip process (not caring whether 3nm or 5nm), coupled with physical laws (light-speed latency, energy conservation, bandwidth ceilings cannot be broken).

Three-layer decoupling:

1. **Interface decoupling:** agents declare compute/memory/model capabilities through a unified capability declaration (Agent Manifest), not caring about the bottom-layer hardware model.
2. **Supply decoupling:** agents can run across cloud, edge, terminal, and browser, with hardware suppliers replaceable.

3. **Cost decoupling:** agents do not own hardware but can buy or contribute compute on demand.

Three-layer coupling:

1. **Physical coupling:** latency (cross-border RTT tens to hundreds of ms), bandwidth ($T = \text{Size} / \text{Bandwidth}$), compute ($T = \text{Compute} / \text{EffectiveFLOPs}$), storage, energy ($E = \text{Power} \times \text{Time}$), thermal throttling.
2. **Economic coupling:** unit-task returns must cover compute + verification + coordination + risk costs (see Section 3.1).
3. **Security coupling:** verification, encryption, TEE, zkML all have extra compute and communication overhead.

Hardware is a multiplier, not an irrelevant item: the swarm-intelligence formula can be rewritten as:

$$C = \frac{C_{\text{cognitive}} \times D_{\text{collab}} \times V_{\text{eff}} \times I_{\text{align}} \times G_{\text{trust}}}{C_{\text{coord}} + C_{\text{verify}} + C_{\text{hardware}} + C_{\text{security}}} \times H_{\text{bonus}}$$

where H_{bonus} is the hardware-dividend multiplier — stronger edge models, larger memory, faster networks, cheaper storage, cheaper electricity all raise H_{bonus} and thus C . Hardware progress is not an internal variable of the P2P network, but it is an external multiplier.

Practical significance of this proposition: designers of P2P agent networks need not (and should not) solve bottom-layer engineering problems like chip processes, memory bandwidth, and electricity materials — these are the hardware industry's business. But designers must **perceive hardware constraints:** task matching needs to consider nodes' compute/memory/network profiles, latency-sensitive tasks need scheduling to local/edge nodes, and energy-sensitive tasks need scheduling to idle-time/low-price nodes. The hardware-abstraction layer (see Section 5.2) is designed precisely for this.

4.5 Falsifiable conditions and forecasts

A scientific hypothesis must be falsifiable. This paper's swarm-intelligence scaling-law hypothesis gives the following clear falsifiable conditions:

Falsifiable condition 1 (verification-cost death sentence, given in Section 3.1): if $C_{\text{verify}} + C_{\text{coord}} > C_{\text{central}} - C_{\text{exec}}$, P2P has no economic viability.

Falsifiable condition 2 (homogenization death sentence, given in Section 3.2): if the entropy of agent capability distribution falls over time and task quality does not grow with N_{eff} , then $\alpha \rightarrow 0$ and scale expansion has no dividend.

Falsifiable condition 3 (D_{collab} falls rather than rises): if as protocols improve and nodes increase, D_{collab} does not rise (i.e. coordination + verification costs grow faster than successful task counts), then "P2P scaling" is only a slogan and the hypothesis is falsified.

Falsifiable condition 4 (N_{eff} ceiling): if N_{eff} stops growing after reaching some threshold (like 100,000) (retention collapse, free-riding too high, Sybil attacks rampant), then the million/billion-level scaling-law extrapolation fails.

Falsifiable condition 5 (governance capture): if G_{trust} falls over time (governance power concentrates, rules manipulated by large nodes), then C 's growth is unsustainable and the network finally degenerates to a centralized platform.

Scoring-time forecast list (design proposal • to implement, [RESULT NEEDED]):

Forecast ID	Forecast content	Scoring time	Verification method	Current status
F1	By end-2027, no production-grade P2P agent collaboration network with $N_{\text{eff}} > 100,000$ exists (inference markets like Bittensor do not count as "collaboration networks")	2027-12-31	Search public data and industry reports	Pending
F2	The unit cost of P2P AI inference (including verification) will not fall below 50% of centralized cloud APIs before 2028	2028-06-30	Independent cost-comparison tests	Pending
F3	zkML cannot support production-grade inference verification of $> 7B$ parameter models before 2028	2028-12-31	Track zkML framework progress	Pending
F4	Any pure P2P (no centralized coordination) agent network has task success rate $< 80\%$ in latency-sensitive scenarios ($< 500\text{ms}$)	2027-06-30	Benchmarking	Pending
F5	The median DAO governance voting participation remains $< 15\%$ over the next 3 years	2029-09-21	Track DAO governance data	Pending
F6	Hybrid architecture (centralized coordination + decentralized execution) will become the mainstream form of P2P agent networks, and pure P2P will be eliminated for insufficient performance	2028-12-31	Industry architecture observation	Pending

Chapter summary

This paper's proposed swarm-intelligence scaling-law hypothesis is falsifiable: C depends on the joint action of N_{eff} , D_{collab} , V_{eff} , I_{align} , and G_{trust} , $\alpha < 1$ (sublinear), and D_{collab} and V_{eff} are decisive variables. The hardware-decoupling boundary proposition clarifies the design principle of "decoupled from implementation, coupled with constraints." Five falsifiable conditions and six scoring-time forecasts make the hypothesis accept future empirical testing. Currently all exponents ($\alpha, \beta, \gamma, \delta, \epsilon$) are [RESULT NEEDED], needing calibration in the UDOS simulation environment or real testnets.

5 Protocol design: eight-layer stack, four protocol objects, verification hierarchy, and dual-track economy

This chapter is the engineering core of the whole paper. Based on the essence of Chapter 2 and the defect constraints of Chapter 3, it proposes the complete protocol design for P2P agent networks. All designs are marked "design proposal • to implement" + [RESULT NEEDED], not claiming implemented or verified.

5.1 The eight-layer protocol stack

Layer	Name	Core function	Key technologies	Design points
L8	Governance layer	Rule-making, dispute arbitration, protocol upgrades, public-fund allocation	DAO, reputation-weighted voting, delegated voting, expert committees, multi-chamber system	Capital can buy compute but not governance rights; can exit and fork
L7	Settlement layer	Contribution metering, task payments, staking slashing, cross-community settlement	Credit (non-transferable), Token (compliant), fiat/stablecoin, smart contracts	Credit first, Token compliant, no appreciation promise
L6	Verification layer	Execution-result verification, malicious detection, dispute handling	Sampling verification, redundant computation, TEE, zkML, optimistic verification + challenge, arbitration	Five-level hierarchy, choosing strength by task value and privacy
L5	Data layer	Data storage, privacy computing, data authorization and returns	IPFS/content addressing, encrypted storage, federated learning, differential privacy, TEE, zero-knowledge proofs	Data not leaving the domain first, local computing, only returning proofs
L4	Execution layer	Task execution, model inference, tool calls, sandbox isolation	WASM, containers, sandboxes, local models, API models, MCP tool calls	Hardware abstraction, heterogeneous scheduling, power awareness
L3	Communication layer	Agent messaging, streaming, NAT traversal	libp2p, WebRTC, QUIC, MCP/A2A compatible	Async first, relay backstop, not requiring all nodes online simultaneously
L2	Discovery layer	Agent capability registration, task indexing, semantic routing	Kademlia DHT, capability routing, agent registry, sharded subnets, bridge nodes	$O(\log N)$ lookup, semantic matching, local markets + cross-domain bridging
L1	Identity layer	Node identity, key management, reputation profiles, contribution history	DID, public keys, verifiable credentials (VC), non-transferable reputation, staking	Identity bound to reputation, reputation non-transferable, anti-Sybil

Hardware-abstraction layer (cross-layer): device-capability registration, task-profile matching, runtime sandboxes, resource metering, execution proofs. The hardware-abstraction layer is not an independent layer but a cross-cutting concern running through L1-L8 — each layer needs to perceive hardware constraints (see Section 4.4).

Figure 2: P2P agent network eight-layer protocol stack + hardware-abstraction layer (cross-cutting)

Figure 2 Conceptual schematic, not measured. The eight-layer protocol stack bottom-up: identity (L1) → discovery (L2) → communication (L3) → execution (L4) → data (L5) → verification (L6) → settlement (L7) → governance (L8); the hardware-abstraction layer is a cross-cutting concern. All layers are design proposals • to implement [RESULT NEEDED].

Design principles:

1. **Not replacing MCP/A2A, but compatible and extending:** the L3 communication layer is compatible with MCP/A2A message formats at the application layer and uses libp2p to replace centralized HTTP at the transport layer.
2. **Async first:** not requiring all agents online simultaneously, tasks can queue and nodes can asynchronously claim (analogous to BOINC's volunteer-computing model).
3. **Local first:** latency-sensitive and privacy tasks scheduled first to local/edge nodes.
4. **Hybrid architecture:** early allowing centralized coordination (the L2 discovery layer can be assisted by centralized indexing services), gradually decentralizing.
5. **Can exit and fork:** any participant can exit the network carrying identity and reputation, and communities can fork the protocol.

5.2 The four core protocol objects

The minimal closed loop of the protocol consists of four objects: **Agent Manifest** → **Task Envelope** → **Receipt** → **Reputation**. These four objects correspond to the four stages of the task lifecycle: discovery → execution → verification → settlement.

Figure 3: Agent magnet-link task closed-loop sequence: four protocol objects and nine-step lifecycle

Figure 3 Conceptual schematic, not measured. The four protocol objects (Agent Manifest/Task Envelope/Receipt/Reputation) correspond to the four stages of the task lifecycle; the nine-step closed loop: discovery → publish → match → take → submit → verify → settle → reputation update → cycle. The JSON schema is a design draft • to implement [RESULT NEEDED].

5.2.1 Agent Manifest (agent magnet link)

The Agent Manifest is the agent's capability declaration and identity card, content-addressed as `agent://<cid>?v=1`, like a BitTorrent magnet link. Anyone obtaining this Manifest can discover, call, assign tasks, provide compute, verify results, and share/seeding.

```
{
  "agent_id": "did:key:z6Mk...",
  "version": "1.0",
  "name": "translation-zh-en",
  "capabilities": ["translate", "summarize", "code_review"],
  "io_schema": {
    "input": {"type": "text", "max_tokens": 8192},
    "output": {"type": "text", "max_tokens": 8192}
  },
  "model_hashes": ["sha256:abc123..."],
  "runtime": ["wasm", "docker", "browser", "api"],
  "hardware_profile": {
    "min_memory_gb": 8,
    "preferred_compute": ["gpu", "npu", "cpu"],
    "max_latency_ms": 5000
  },
  "tools": ["ocr", "glossary", "web_search"],
  "pricing": {
    "unit": "1k_tokens",
    "amount": "0.01",
    "currency": "credit"
  },
  "verification": {
    "default_mode": "sample",
    "supported_modes": ["sample", "redundant", "tee", "zkml"],
    "challenge_window": "1h"
  },
  "stake": {"amount": 100, "currency": "credit"},
  "reputation": "did:key:z6Mk.../rep",
  "endpoints": ["libp2p://Qm...", "https://..."],
  "dht_key": "sha256:def456...",
  "settlement": "chain:0x...",
  "license": "Apache-2.0",
  "privacy_policy": "data_not_stored",
  "signature": "..."
}
```

Key-field notes:

- `agent_id`: DID identity, bound to the public key, unforgeable.
- `capabilities` + `io_schema`: capability declaration, used for DHT semantic routing and task matching.
- `model_hashes`: declares the model-weight hashes used, and verifiers can confirm execution used the declared model.
- `hardware_profile`: hardware profile (see Section 4.4), used for power-aware scheduling.
- `verification`: declares supported verification modes, and task publishers can choose appropriate verification strength.

- `stake`: staking amount, slashed on malicious acts.
- `reputation`: reference pointing to the reputation profile, reputation non-transferable.

5.2.2 Task Envelope

The Task Envelope is the complete definition of the task, including input, verification strategy, bounty, privacy requirements, and deadline.

```
{
  "task_id": "cid:task789...",
  "type": "translation",
  "input_cid": "cid:bafy...",
  "input_hash": "sha256:...",
  "output_requirements": {
    "type": "text",
    "quality_bar": "human_parity",
    "max_tokens": 4096
  },
  "verification": {
    "mode": "redundant",
    "redundancy": 3,
    "consensus_threshold": "2_of_3",
    "challenge_window": "24h"
  },
  "privacy": {
    "level": "local_only",
    "data_retention": "delete_after_verification",
    "allowed_regions": ["CN", "SG"]
  },
  "budget": {
    "execution_fee": 10,
    "verification_fee": 3,
    "protocol_fee": 0.5,
    "currency": "credit"
  },
  "deadline": "2026-09-22T00:00:00Z",
  "shard": {"total": 1, "index": 0},
  "publisher": "did:key:z6Mk...",
  "signature": "..."
}
```

Isomorphic relation with UDOS Paper 09's TransferBundle: the Task Envelope is isomorphic to the TransferBundle's six fields (Goal/Context/Done/Trace/Owner) — Goal→task definition, Context→input CID and privacy requirements, Done→output requirements and verification strategy, Todo→execution and deadline, Trace→resource metering and Receipt, Owner→publisher signature.

5.2.3 Receipt

The Receipt is the proof of the execution result, including result hash, resource metering, signature, and optional TEE/zk proofs.

```
{
  "receipt_id": "cid:receipt012...",
  "task_id": "cid:task789...",
  "executor": "did:key:z6Mk...",
  "result_hash": "sha256:result...",
  "result_cid": "cid:bafyresult...",
  "resource_metering": {
    "compute_flops": 1.5e12,
    "memory_peak_gb": 6.2,
    "network_bytes_in": 10240,
    "network_bytes_out": 20480,
    "duration_ms": 3200,
    "energy_wh": 0.0045
  },
  "model_used": "sha256:abc123...",
  "proof": {
    "type": "tee_attestation",
    "evidence": "base64...",
    "verified_by": "did:key:verifier..."
  },
  "verification_result": {
    "mode": "redundant_2_of_3",
    "passed": true,
    "verifiers": ["did:key:v1...", "did:key:v2...", "did:key:v3..."],
    "dispute": null
  },
  "timestamp": "2026-09-21T12:00:00Z",
  "signature": "..."
}
```

Resource metering is the basis of settlement: the resource_metering field in the Receipt precisely records the compute, memory, network, time, and energy consumed by execution, the basis of contribution metering and settlement. This solves the core problem of "contributions not measurable" in volunteer computing.

5.2.4 Reputation

Reputation is the multidimensional reputation aggregated from Receipts, non-transferable, determining priority, staking quota, and governance weight.

```
{
  "agent_id": "did:key:z6Mk...",
  "dimensions": {
    "quality": 0.92,
    "speed": 0.85,
    "honesty": 0.98,
    "availability": 0.78
  },
  "aggregate": 0.89,
  "history": {
    "total_tasks": 1843,
    "successful": 1712,
    "disputed": 23,
    "slashed": 5
  },
  "decay": {
    "half_life_days": 90,
    "last_update": "2026-09-21"
  },
  "stake_amount": 100,
  "governance_weight": 0.89,
  "signature": "..."
}
```

Design points:

- **Multidimensional reputation:** four dimensions of quality, speed, honesty, availability, and different tasks can weight different dimensions (e.g. high-value tasks weight honesty and quality, batch tasks weight speed and availability).
- **Non-transferable:** reputation bound to the DID identity, not buyable/sellable. This prevents capital from obtaining governance rights by buying reputation — "capital can buy compute, but cannot buy reputation."
- **Decay mechanism:** reputation decays over time (90-day half-life), preventing early contributors from "lying flat" on old credit and also reducing the long-term returns of Sybil reputation farming.
- **Staking association:** the higher the reputation, the lower the required staking (because malicious probability is low); the lower the reputation, the higher the required staking. This aligns reputation with economic incentives.

5.3 Five-level verification hierarchy

Based on the analysis in Section 2.3 (zkML performance poor, TEE hardware root of trust, optimistic verification latency high), no single verification scheme covers all scenarios. This paper proposes a five-level verification hierarchy, choosing appropriate strength by task value and privacy requirements:

Level	Name	Verification method	Verification overhead	Finality delay	Applicable scenarios	Malicious cost
L0	Self-report + sampling	Executor self-reports results, randomly sampling 5-10% to re-verify	Extremely low (5-10% execution cost)	Immediate (after sampling)	Low-value, low-risk tasks (translation, summary, annotation)	Sampling-detection probability $\times$ slashing
L1	Redundant 2-of-3	3 independent nodes execute, 2/3 agreement passes	Medium (about 200% execution cost)	Minutes (waiting for 3 nodes)	Medium-value tasks (code review, data analysis, document Q&A)	Need to control 2/3 nodes
L2	TEE trusted execution	Executor executes in the TEE, submits remote attestation	Low (TEE performance overhead 5-15%)	Immediate (attestation verification fast)	Privacy-sensitive tasks (medical, financial, personal data)	Need to breach TEE side channels
L3	zkML + optimistic challenge	Executor submits zk proof, sets challenge window	High (zk proof generation 100-1000 $\times$ execution time)	Challenge window (hours to days)	High-value, formally verifiable tasks	Need to breach the zk proof system
L4	Committee arbitration + staking slashing	Multisig committee reviews disputes, staking-slashes malicious actors	High (human/committee review)	Days to weeks	Disputed tasks, high-value disputes, protocol upgrades	Reputation destruction + economic slashing

Economic model of verification cost: the choice of verification strength should satisfy:

$$C_{\text{verify}}(\text{level}) \leq V_{\text{task}} \times P_{\text{fraud}}$$

i.e. verification cost does not exceed task value times fraud probability. For a \$0.01 translation task, L0 sampling suffices; for a \$10,000 medical-diagnosis task, L2 TEE + L4 arbitration is needed.

Mapping with UDOS Paper 06: the five-level verification hierarchy is isomorphic to Paper 06's three-level typed finality (probabilistic / economic / deterministic) — L0 corresponds to probabilistic finality (sampling passes means probably correct), L1-L3 to economic finality (malicious cost > return means rational non-malice), L4 to deterministic finality (irreversible after committee arbitration).

5.4 Dual-track economic model: credit, Token, and fiat

Three-track separation principle (compliance red line):

Track	Nature	Transferability	Use	Compliance requirement
Credit	Internal contribution-accounting unit	Non-transferable/non-tradeable	Reputation, quotas, priority, staking	None (no monetary value)
Token (if needed)	Cross-community settlement and staking	Restricted transfer	Cross-community settlement, staking, governance	Compliant issuance, no appreciation promise, not investment advice
Fiat/stablecoin	Real payment	Fully transferable	Enterprise procurement, wages, tax	KYC/AML, licensed channels

Why is credit non-transferable? If credit were transferable, it would have monetary value and might be determined by regulators as a security or virtual currency, triggering compliance risk. Non-transferable credit is purely a "contribution proof," like airline miles (the non-transferable version), used to raise service levels rather than trade.

Contribution-metering formula (design proposal • to implement):

$$Contribution = \sum_i (Resource_i \times Time_i \times Utilization_i \times Quality_i \times Scarcity_i) - Penalty$$

where:

- Resource_i: the resource amount contributed (compute FLOPs, storage GB, bandwidth MB, data volume, feedback count).
- Time_i: contribution duration.
- Utilization_i: resource utilization (preventing false reporting of idle but actually not contributing).
- Quality_i: quality coefficient (from Reputation's quality dimension, 1 if Receipt verification passes, 0 or negative if fails).
- Scarcity_i: scarcity coefficient (scarce resources like GPU compute have higher coefficients than abundant resources like CPU compute).
- Penalty: punishment (malicious slashing, task failure, dispute loss).

Distribution principle: automatically settle by Receipt; disputed tasks delay settlement until arbitration ends; the public pool (a certain proportion of protocol fees, suggested 10%) is used for cold-start subsidies, security audits, and public goods (open-source models, datasets, verification tools).

Anti-capital-monopoly design:

- Capital can buy compute (contributing more Resource_i), but cannot buy Quality_i (quality comes from historical performance) and Reputation (non-transferable).
- Governance weight is based on Reputation (reputation weighting) rather than Token holdings (token weighting), preventing whales from controlling governance.
- Delegated voting lets small contributors delegate governance to trusted representatives, but representatives must periodically disclose voting records.
- Multi-chamber governance: contributor committee (reputation-weighted) + expert committee (technical capability) + user committee (user interests) + public-fund oversight committee, with any protocol upgrade needing multi-chamber co-signature.

5.5 Role differentiation: not everyone runs large models

Nodes in P2P agent networks are not homogeneous but differentiated into different roles:

Role	Contribution	Typical hardware	Return source	Ordinary-person participation
Compute node	Inference, verification, storage	Phones, PCs, edge devices	Compute-contribution credit	High (idle-time computing while phone charging)
Model node	Expert models, tools, algorithms	GPU servers	Model-use fees	Medium (needs technical capability)
Data node	Private data (federated learning + differential privacy + TEE)	Phone/PC	Data-authorization returns	High (data not leaving the domain)
Agent node	Personal agent, negotiating/executing/collaborating on one's behalf	Phone/PC	Task-execution fees	High (everyone has an agent)
Verification node	Sampling audits, redundant computation, challenge arbitration	PC/server	Verification fees	Medium (needs reputation and staking)
Routing node	Task decomposition, matching, cross-subnet bridging	Servers	Routing fees	Low (needs technical capability)
Governance node	Rule-making, dispute arbitration, public funds	No special hardware	Governance stipends	Medium (needs high reputation)
Compliance/insurance node	Legal entities, insurance pools, audits, liability isolation	No special hardware	Service fees	Low (needs legal/financial qualification)

Key insight: the core of ordinary people's contribution is not idle compute. The market value of idle compute is very low — the marginal cost of global GPU compute is priced by centralized cloud, and P2P idle compute is hard to compete on price. Ordinary people's truly scarce contributions are:

1. **Long-tail scenarios and local knowledge:** localized, verticalized, real-time scenarios centralized platforms cannot cover (like community Q&A, local experience, minority-language translation).
2. **Data sovereignty:** the authorized use of personal data under privacy protection, which centralized platforms cannot legally obtain.
3. **Human feedback:** annotation, evaluation, correction, preferences — these are core resources of RLHF and model evaluation and cannot be replaced by compute.
4. **Verification and governance:** human review, dispute arbitration, community governance — these need human judgment and cannot be automated.
5. **Social trust:** acquaintance networks, community reputation, guarantee relationships — this is the social capital of anti-Sybil.

Chapter summary

The eight-layer protocol stack (identity → governance) defines the complete architecture of P2P agent networks; the four protocol objects (Agent Manifest/Task Envelope/Receipt/Reputation) form the minimal closed loop of the task lifecycle; the five-level verification hierarchy chooses verification strength by task value and privacy; the dual-track economy (non-transferable credit + compliant Token +

KYC fiat) achieves incentive alignment within compliance red lines; role differentiation clarifies that ordinary people's core contribution is not idle compute but long-tail scenarios, data sovereignty, feedback, verification, and social trust. All designs are "design proposals • to implement" [RESULT NEEDED].

| 6 Mapping with the existing UDOS system

This chapter establishes the precise mapping of P39 with the 13 papers of the UDOS v7.7.2 master volume, ensuring the supplementary volume does not contradict the existing system but extrapolates and deepens on its basis.

6.1 Paper-by-paper mapping table

UDOS paper	Theme	P39's mapping and extrapolation	Relation
Paper 02	Structure×scale scaling law, three-segment curve, paradigm wall	P39 supplements "network structure" as the decentralized extrapolation of T (connection topology) in the structural triplet $S=(T,G,M)$; the distinction between N_{eff} and the scale axis N; P2P's "paradigm wall" = verification-cost wall	Extrapolation and deepening
Paper 04	RSI convergence path, L1-L5 lineage	Whether P2P networks constitute a mass line of "structural self-modification" — agent networks self-reorganizing topology = distributed L4/L5? Cautious judgment: no existing P2P network reaches RSI self-modification, marked [RESULT NEEDED]	Cautious extrapolation
Paper 05	Coordination topology, hierarchical topology, communication protocols, memory wall, million-level extrapolation	P2P's DHT/semantic routing/sharded subnets = decentralized implementation of fan-in $O(\log N)$; local markets + bridge agents = hierarchical hybrid tree; Paper 05's memory-complexity dimension (Warp-Cortex $O(N \cdot L) \rightarrow O(1)$ weights) corresponds in P2P to distributed sharding of state storage	Direct extrapolation
Paper 06	Consensus stopping, BFT-lite, three-level typed finality, rules first	Receipt + layered verification = decentralized BFT-lite; verification cost = marginal cost of the $2f+1$ quorum; five-level verification (L0 sampling $\rightarrow$ L4 arbitration) is isomorphic to three-level finality (probabilistic/economic/deterministic); The Cost of Consensus's sycophantic compliance is the theoretical basis of P2P homogeneous saturation	Direct extrapolation
Paper 07	Quality saturation law, heterogeneity dividend, simulation validity	Isomorphic copies bring no gain — P2P node heterogeneity/independent information sources are the dividend; Bittensor-style homogeneous subnets = homogeneous-expansion saturation; the quality saturation law $mse(k)=a+b/k$ directly applies to choosing the optimal copy count for P2P redundant verification	Core theoretical basis
Paper 09	Handoff contract, TransferBundle, MCP/A2A interop	Task Envelope/Receipt is isomorphic to the TransferBundle six fields (Goal $\rightarrow$ task definition, Context $\rightarrow$ input CID, Done $\rightarrow$ output requirements, Todo $\rightarrow$ execution and deadline, Trace $\rightarrow$ resource metering, Owner $\rightarrow$ signature); the docking of Agent Manifest with A2A Signed Agent Card; the adaptation of MCP stateless core at the P2P transport layer	Protocol-object isomorphism
Paper 10	Agent task market, settlement conservation, Byzantine slashing, VCG, settlement neutrality	UDOS's internal task market (budget/slashing/conservation balance_sum=total_paid) directly extrapolates to the P2P settlement market; open market vs closed accounting; the application of VCG-style payments in P2P task bidding; the challenge of settlement neutrality in decentralized settlement	Direct extrapolation
Paper 13	Evidence engineering, evidence grading, literature ledger	Figures of token projects like Bittensor/io.net uniformly three-state; not endorsing any project; P39's EVIDENCE_LEDGER follows Paper 13's evidence-grading standards	Methodological consistency

6.2 Systematic dialogue with Paper 02's scaling law

Paper 02 established the structure×scale three-segment scaling-law curve: linear growth → sublinear growth → paradigm wall (verified, tracing reports7/). P39's swarm-intelligence scaling-law hypothesis is the concrete unfolding of this framework on the "network-structure axis":

- **First segment (linear growth):** when N_{eff} is small (<1000), coordination and verification costs are low, and adding nodes approximately linearly raises C .
- **Second segment (sublinear growth):** when N_{eff} grows to 10,000-100,000, coordination and verification costs begin eating scale returns, $\alpha < 1$.
- **Third segment (paradigm wall):** when N_{eff} exceeds some threshold, the growth of verification + coordination costs exceeds execution-cost savings, and C peaks or even falls. This threshold is P2P's "paradigm wall" — determined not by model capability but by protocol efficiency (D_{collab} and V_{eff}).

Key distinction: Paper 02's paradigm wall is mainly determined by hardware/memory/communication physical constraints; P39's paradigm wall is mainly determined by protocol efficiency and verification cost. This means P2P's paradigm wall is **movable through protocol improvement** — if D_{collab} and V_{eff} rise with protocol improvement, the paradigm-wall threshold can be raised. This is precisely the meaning of falsifiable condition 3 (if D_{collab} falls rather than rises, the hypothesis is falsified).

6.3 Cautious dialogue with Paper 04's RSI

Paper 04 established RSI's L1-L5 lineage (verified). A natural question is: do P2P agent networks constitute "distributed RSI"?

Cautious judgment:

- **L1-L2 (tool use/parameter self-tuning):** a single agent in a P2P network can reach L1-L2, but this is not network-level RSI.
- **L3 (architecture self-modification):** if the P2P network's protocol itself can automatically upgrade through governance voting (like a DAO voting through new verification rules and automatically deploying), this may constitute network-level L3. But currently no P2P network has implemented this [RESULT NEEDED].
- **L4-L5 (goal self-modification/complete self-evolution):** the P2P network's goals are determined by governance, and if governance can modify governance rules themselves (amending the constitution), this may constitute L4. But this brings huge security risk — "modifying the rules that modify rules" may cause value drift and uncontrollable evolution. Paper 04 has warned of RSI's convergence risk, and P2P networks' distributed RSI risk is greater (no single entity can emergency-stop).

Conclusion: the relation of P2P agent networks to RSI is "potential carrier" rather than "implemented." Currently all P2P networks fail to reach network-level RSI, marked [RESULT NEEDED]. In design, the P2P network's governance layer should set "constitutional rigid clauses" — certain core rules (like privacy protection, exit rights, anti-Sybil mechanisms) cannot be modified by ordinary governance votes, needing higher thresholds (like a full referendum + delayed effectiveness + rollback mechanisms) to prevent value drift.

Chapter summary

P39 forms a precise mapping with the 13 UDOS papers: Paper 02 provides the scaling-law framework, Paper 05 the coordination-topology basis, Paper 06 the consensus and verification theory, Paper 07 the core basis of quality saturation and heterogeneity dividends, Paper 09 the isomorphism basis of protocol objects, Paper 10 the task-market and settlement theory, Paper 13 the evidence methodology. The relation to Paper 04's RSI is a cautious "potential carrier" rather than implemented. P39 is not a new paper independent of the UDOS system but the extrapolation and deepening of the existing system on the P2P agent-network dimension.

7 Cold-start path and hybrid architecture

7.1 Why not start with a "global billion-level P2P"

P2P networks have the classic cold-start problem: network value is proportional to the square of node count (Metcalfe's law), but early when nodes are few network value is low, unable to attract new nodes. Moreover:

- **Pure P2P performance unreachable:** as described in Section 3.6, pure P2P is far inferior to centralization in latency, reliability, and SLA. If starting with pure P2P, user experience is poor, retention low, and the network cannot grow.
- **Verification infrastructure missing:** the five-level verification hierarchy needs infrastructure like TEE, zkML, and arbitration committees, impossible to all be ready early.
- **Governance mechanisms need running-in:** DAO governance, reputation systems, and dispute arbitration need to iterate in practice and cannot be perfect from the start.
- **Compliance risk:** introducing Token and cross-border settlement early makes compliance complexity extremely high and may cause the project to be halted by regulators.

Therefore, **the hybrid architecture (centralized coordination + decentralized execution) is the only engineering-feasible cold-start path.**

7.2 Six-stage cold-start roadmap

Stage	Name	Node scale	Architecture	Core task	Key milestone	Decentralization degree
0	Single-machine client	1	Pure centralization	Personal agent management, running models locally, connecting APIs	Client usable, local inference working	0% (pure centralization)
1	Invitation-only small community	100-1000	Centralized coordination + decentralized execution	Sharing compute/ models/agents, running credit, human verification	100-node closed loop, task success >80%	10% (execution decentralization)
2	Agent magnet links	1000-10000	Centralized indexing + P2P discovery	Agent Manifest standardization, DHT discovery, one-click share/ run/seeding	1000 agents discoverable and callable	30% (discovery decentralization)
3	Vertical task market	10,000-100,000	Centralized matching + P2P execution + automatic verification	Task publishing/ taking/bidding, sampling verification, reputation, credit settlement	Daily 10,000 tasks, verification cost <50% execution	50% (execution + verification decentralization)
4	Cross-platform protocols	100,000-1,000,000	Sharded DHT + local markets + bridge nodes	MCP/A2A compatible, federated subnets, Token/ stablecoin settlement, dispute arbitration	Cross-3-community interop, settlement automated	70% (network + settlement decentralization)
5	Hybrid open network	1,000,000-1,000,000,000	Centralized coordination optional + full-stack decentralization	Sharded DHT million agents, DAO governance, public funds, insurance compliance	N_eff>100,000, governance participation >15%	90% (governance decentralization, coordination still optionally centralized)

Figure 4: P2P agent network six-stage cold-start roadmap and hybrid architecture

Figure 4 Conceptual schematic, not measured. Six stages: single-machine client (0%) → invitation-only small community (10%) → agent magnet links (30%) → vertical task market (50%) → cross-platform protocols (70%) → hybrid open network (90%). The hybrid architecture is a permanent choice

(centralized coordination optional + P2P network layer + local execution layer), not an expedient. Each stage's scale and decentralization degree are design targets • to implement [RESULT NEEDED].

Core-scenario entry for each stage:

- **Stages 0-1:** personal knowledge management, local document Q&A, lightweight agent assistants (not relying on the network).
- **Stages 2-3:** translation, data annotation, code review, content moderation, simulation compute (splittable, verifiable, latency-insensitive).
- **Stages 4-5:** scientific reproduction, long-tail knowledge Q&A, federated learning, multi-agent collaboration, decentralized AI service markets.

7.3 Design principles of the hybrid architecture

The hybrid architecture is not an expedient of "temporarily centralized, future decentralized" but a **permanent architecture choice**. Even at stage 5, centralized coordination remains optional — for low-latency, high-SLA tasks users can choose a centralized coordinator (paying); for cost-sensitive, latency-insensitive tasks users can choose pure P2P.

Three layers of the hybrid architecture:

1. **Centralized coordination layer (optional):** task indexing, matching, scheduling, quality monitoring. Operated by a foundation or compliant company, charging service fees. Users can choose to use or skip.
2. **P2P network layer:** agent discovery, communication, execution, verification, settlement. Fully decentralized, no single point of failure.
3. **Local execution layer:** agent runtime, local models, private data on personal devices. Fully local, data not leaving the domain.

Why is pure P2P not the goal? As described in Chapter 3, pure P2P has inherent defects in performance, reliability, and SLA. The hybrid architecture acknowledges these defects, uses centralized coordination to make up P2P's shortcomings, and uses P2P to make up centralization's monopoly and censorship risks. **The goal is not "decentralization" but "choice"** — users can freely choose between centralization and decentralization according to task needs.

7.4 Key metrics and success criteria

Metric	Definition	Stage 3 target	Stage 5 target	Failure criterion
N_eff	Active nodes continuously contributing and passing the reputation threshold	>1000	>100,000	<5% of registered nodes
Task success rate	Proportion of tasks passing verification and accepted by the demand side	>85%	>95%	<70%
Verification-cost ratio	$C_{\text{verify}} / C_{\text{exec}}$	<50%	<20%	>100% (verification death sentence)
Unit task cost	End-to-end cost including verification + coordination	<150% of centralized API	<80% of centralized API	>200% of centralized API
Free-riding rate	Proportion of nodes only consuming not contributing	<40%	<25%	>70%
Governance participation	Participation in governance voting	>5%	>15%	<1% (governance capture)
Ordinary-person return share	Proportion of returns obtained by non-institutional nodes	>30%	>50%	<10% (centralization resurgence)
Privacy-incident rate	Data-leak/privacy-violation events per quarter	<1	0	>5/quarter

Core success criterion: at stage 5, verification-cost ratio <20% and unit task cost <80% of centralized API and ordinary-person return share >50%. If these three conditions simultaneously hold, the P2P agent network's economic rationality and the "mass line's" value promise simultaneously hold. If any condition fails, return to protocol-design improvement or acknowledge P2P inapplicable in that scenario.

Chapter summary

Cold start adopts the six-stage roadmap (single machine → small community → magnet links → task market → cross-platform → hybrid open network), with decentralization degree gradually rising. The hybrid architecture is a permanent choice rather than an expedient — centralized coordination makes up P2P's performance defects, P2P makes up centralization's monopoly risk, and the goal is "choice" rather than "pure decentralization." Eight key metrics define the success criteria, among which verification-cost ratio, unit task cost, and ordinary-person return share are core.

8 Risks, failure boundaries, falsifiable conditions, and forecasts

This chapter aggregates the whole paper's risk list, failure boundaries, falsifiable conditions, and forecasts, the paper's critical closure.

8.1 Risk panorama table

Risk ID	Risk	Severity	Likelihood	Impact	Mitigation	Residual risk
R1	Verification-cost death sentence	Fatal	Medium	P2P no economic viability, architecture fails	Five-level layered verification, L0 sampling first, only high-value tasks use high-cost verification	Medium (generative-task verification essentially hard)
R2	Homogeneous saturation	High	High	Scale expansion no dividend, $\alpha \rightarrow 0$	Incentivize heterogeneity, reward independent information sources, punish convergence	Medium (verification mechanisms may naturally incentivize convergence)
R3	Sybil attack	High	High	Reputation/governance/matching manipulated	DID + non-transferable reputation + staking + Proof of Personhood + behavior analysis	Medium (no perfect anti-Sybil scheme)
R4	Free-riding/tragedy of the commons	High	High	Supply side shrinks, network degenerates	Contribution metering + credit consumption + priority tilt + staking	Medium (free-riding is P2P's inherent problem)
R5	Governance capture/whale control	High	Medium	Decentralization promise bankrupt, degenerates to centralization	Reputation weighting rather than token weighting + multi-chamber system + delegated voting + constitutional rigid clauses	Medium (information asymmetry makes elite control hard to avoid)
R6	Centralization resurgence	High	High	Entry/compute/model/governance re-centralized	Can exit and fork + multiple clients + open protocols + anti-monopoly design	High (network effects naturally favor centralization)
R7	Token speculation/compliance risk	High	Medium	Project halted by regulators, community split	Credit first, Token compliant, no appreciation promise, KYC/AML	Medium (regulatory environment uncertain)
R8	Privacy leak/data security	High	Medium	User data leaked, legal liability	Local first + encryption + differential privacy + TEE + data not leaving the domain	Medium (TEE has side channels, federated learning has gradient leaks)
R9	Prompt injection/data poisoning/model theft	Medium	High	Execution results untrusted, model IP lost	Sandbox isolation + input filtering + redundant verification + model watermarking + reputation punishment	Medium (agent security is an open question)
R10		Medium	High			High (P2P's physical latency)

Risk ID	Risk	Severity	Likelihood	Impact	Mitigation	Residual risk
	Performance inferior to centralization			User experience poor, retention low	Hybrid architecture + local first + edge caching + async first	cannot be eliminated)
R11	Smart-contract security/fund loss	Medium	Medium	Settlement funds stolen	Formal verification + audits + insurance pools + gradual decentralization	Low (mitigable through mature schemes)
R12	Distributed RSI out of control	Fatal	Low	Network-level self-evolution uncontrollable	Constitutional rigid clauses + emergency-stop mechanisms + multisig governance + rollback	Low (current technology far from reaching, but needs pre-design)

8.2 Failure-boundary summary

This paper's propositions and designs hold within the following boundaries, and beyond the boundaries they are inapplicable or need correction:

1. **Scenario boundary:** P2P applies to batch, splittable, verifiable, latency-insensitive tasks; not to low-latency real-time inference, large-model pretraining, strong privacy sensitivity, high-frequency trading, or enterprise services needing SLA (see Section 3.6).
2. **Scale boundary:** the swarm-intelligence scaling law's $\alpha < 1$ holds when $N_{\text{eff}} < 1$ million; whether it still holds when $N_{\text{eff}} > 1$ million [RESULT NEEDED], with new paradigm walls possibly appearing.
3. **Verification boundary:** the five-level verification hierarchy applies to tasks formally or redundantly verifiable; for purely subjective generative tasks (like artistic creation, literary writing), verification is essentially hard and L4 arbitration may be the only scheme.
4. **Economic boundary:** P2P's economic rationality rests on $C_{\text{verify}} + C_{\text{coord}} < C_{\text{central}} - C_{\text{exec}}$; if centralized API prices keep falling (like inference cost falling 50% annually 🟡), P2P's cost advantage may be compressed.
5. **Governance boundary:** multi-chamber governance is effective when participants have basic technical literacy and information access; under severe information asymmetry or participant apathy, governance may degenerate to elite control.
6. **Compliance boundary:** the feasibility of Token and cross-border settlement depends on jurisdictions' regulatory policy; in jurisdictions strictly banning virtual currency, only credit and fiat channels can be used.
7. **Hardware boundary:** P2P network performance is limited by nodes' hardware level; in scenarios with a high proportion of mobile nodes, total compute and network quality may be insufficient to support complex tasks.
8. **RSI boundary:** the relation of P2P networks to RSI is "potential carrier," and currently no P2P network reaches network-level RSI; distributed RSI risk is greater than single-body RSI (no single entity can emergency-stop), needing pre-set constitutional rigid clauses.

8.3 Falsifiable-condition summary

This paper's core hypothesis (swarm-intelligence scaling law) gives five falsifiable conditions (see Section 4.5), summarized as follows:

1. **Verification-cost death sentence:** if $C_{\text{verify}} + C_{\text{coord}} > C_{\text{central}} - C_{\text{exec}}$, P2P has no economic viability.
2. **Homogenization death sentence:** if the entropy of agent capability distribution falls over time and task quality does not grow with N_{eff} , then $\alpha \rightarrow 0$.
3. **D_collab falls rather than rises:** if as protocols improve and nodes increase D_{collab} does not rise, "P2P scaling" is only a slogan.
4. **N_eff ceiling:** if N_{eff} stops growing after some threshold (retention collapse, free-riding too high, Sybil rampant), the million/billion-level extrapolation fails.
5. **Governance capture:** if G_{trust} falls over time, C's growth is unsustainable and the network degenerates to a centralized platform.

Moreover, this paper's design proposals (eight-layer stack, four protocol objects, five-level verification, dual-track economy) are all falsifiable engineering propositions: if after implementation in a testnet the key metrics (task success rate, verification-cost ratio, unit task cost, ordinary-person return share) do not meet targets, the design needs correction or abandonment.

8.4 Forecast list (scoring time)

Six scoring-time forecasts were given in Section 4.5 (F1-F6). Three long-term forecasts are added here:

Forecast ID	Forecast content	Scoring time	Verification method
F7	By 2030, P2P agent networks will occupy >10% market share in at least one vertical scenario (like data annotation or translation)	2030-12-31	Industry market reports
F8	By 2030, no production-grade P2P agent network fully decentralized (without any centralized component) exists; the hybrid architecture will be the only form	2030-12-31	Industry architecture observation
F9	By 2032, the swarm-intelligence scaling law's α exponent will be empirically calibrated in at least one real P2P network, with $0.3 < \alpha < 0.7$	2032-12-31	Academic papers or industry white papers

Chapter summary

12 risks (including 2 fatal), 8 failure boundaries, 5 falsifiable conditions, and 9 scoring-time forecasts together form this paper's critical closure. P2P agent networks are not a universal solution and have clear failure boundaries on eight dimensions: scenario, scale, verification, economy, governance, compliance, hardware, and RSI. Honestly facing these boundaries is the premise of engineering-oriented argument and also the requirement of "taking the essence and removing the defects."

Conclusion: the mass line is not a slogan but an engineering proposition

The core conclusions of this paper can be summarized in three sentences:

First, the bottleneck of million-level agents is coordination rather than single-body intelligence. When agent counts enter the million-to-billion level, the five walls the system meets (coordination, trust,

verification, incentives, governance) are all organizational problems rather than model-capability problems. The large-model scaling law cannot be directly extrapolated to agent networks and needs a new formalization framework.

Second, the scaling law of P2P agent networks is a falsifiable engineering hypothesis, not a revolutionary slogan. This paper proposes $C \approx (N_{\text{eff}})^{\alpha} \times (D_{\text{collab}})^{\beta} \times (V_{\text{eff}})^{\gamma} \times (I_{\text{align}})^{\delta} \times (G_{\text{trust}})^{\epsilon}$ ($\alpha < 1$), clarifies the distinction between N_{eff} and registered node count, emphasizes the decisive role of D_{collab} and V_{eff} , and gives five falsifiable conditions. If verification cost exceeds centralized recomputation, if agent homogenization causes saturation, if D_{collab} does not rise with protocol improvement, then "P2P scaling" is falsified.

Third, the value promise of the "mass line" must land on verifiable mechanism design. Participation descent needs low-threshold light nodes, return descent needs verifiable contribution metering and the dual-track economy (non-transferable credit + compliant Token + KYC fiat), governance descent needs Sybil-resistant reputation systems and multi-chamber governance. The eight-layer protocol stack, four protocol objects, five-level verification hierarchy, and six-stage cold-start roadmap are the concrete designs of these mechanisms (all design proposals • to implement [RESULT NEEDED]). The hybrid architecture (centralized coordination + decentralized execution) is the only engineering-feasible path, and pure P2P is unreachable for performance defects.

Contribution to the UDOS system: P39 extrapolates Paper 02's scaling-law framework to the network-structure axis, Paper 05's coordination topology to DHT/semantic routing/sharded subnets, Paper 06's consensus verification to the five-level verification hierarchy, applies Paper 07's quality saturation law to P2P homogenization critique, makes Paper 09's TransferBundle isomorphic to Task Envelope/Receipt, and extrapolates Paper 10's task market to the open P2P settlement market. P39 is not an independent new paper but the complete extrapolation and deepening of the UDOS system on the P2P agent-network dimension.

Final judgment: P2P agent networks have a real technical basis (libp2p, DHT, MCP/A2A, TEE, federated learning), real demand (anti-monopoly, data sovereignty, long-tail scenarios, inclusive AI), but also real fatal risks (verification-cost death sentence, homogeneous saturation, Sybil attacks, governance capture, centralization resurgence). It is not a revolution bound to succeed but an engineering proposition needing gradual verification in testnets, possibly succeeding in some scenarios and failing in others.

The true meaning of the "mass line" is not "everyone runs large models" but "everyone owns an agent, everyone can contribute, verify, govern, and share the returns of intelligence." Whether this vision can be realized depends not on how loud the slogan is, but on whether protocols can press verification cost low enough, raise collaboration density high enough, and truly keep governance power in ordinary people's hands. This is a falsifiable engineering proposition; let us see the outcome in testnets.

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Evidence discipline and reproduction notes

1. **UDOS's own figures:** the UDOS-system conclusions this paper cites (structural triplet $S=(T,G,M)$, three-segment scaling-law curve, quality saturation law $mse(k)=a+b/k$, heterogeneity dividends, BFT-lite three-level finality, TransferBundle six fields, task-market settlement conservation) all trace back to the raw JSON in release-v5.4.4/udos-engine/reports7/, marked on three levels verified / cpu-proto / cpu-proxy / unverified. This paper adds no new UDOS simulation data, and all UDOS figures follow the v7.7.2 scope.
2. **External literature three states:** the external literature and industry systems this paper cites are marked on three states ✅ verified (with authoritative source URL, accessed 2026-09-21) / 🟡 paper-reported or company-disclosed, not independently reviewed / 🟤 no evidence found. Node counts,

throughput, and unit costs of token projects like Bittensor, Golem, Akash, Render, io.net are all 🟡 (vendor self-report, not independently reviewed), not entering the verified column; but critical data like Akash CPU utilization 0.30%, io.net registered/online $49\times$ gap, Bittensor revenue \$3-15M and $175-400\times$ revenue multiples have independent academic/on-chain sources ✅ BOINC/Folding@home scale figures are ✅ (official + academic sources). RSI-related five-level framework (arXiv:2609.11873), Dream-RSI (arXiv:2609.14858), Gödel Agent (arXiv:2410.04444), HyperAgents (arXiv:2603.19461) are all ✅ (really exist, but all are restricted self-improvement, none end-to-end rewriting weights); "Theseus Labs" institutional attribution 🟤 no evidence found, changed to cite the arXiv original paper.

3. **Design proposals:** this paper's eight-layer protocol stack, four protocol objects (Agent Manifest/ Task Envelope/Receipt/Reputation), five-level verification hierarchy, dual-track economic model, contribution-metering formula, and six-stage cold-start roadmap are all "design proposals • to implement" + [RESULT NEEDED], not claiming implemented or verified. All JSON schemas are design drafts, and fields and values need iteration in testnets.
4. **Scaling-law exponents:** in the swarm-intelligence scaling-law formula this paper proposes, $\alpha, \beta, \gamma, \delta, \epsilon$ are all [RESULT NEEDED], needing calibration in the UDOS simulation environment or real testnets. This paper only presets $0 < \alpha < 1$ (sublinear), not presetting specific values.
5. **Forecasts:** this paper's 9 forecasts (F1-F9) are all scoring-time falsifiable forecasts, current status "pending." Forecasts are based on this paper's theoretical analysis and industry observation, not investment advice or technical promises.
6. **Compliance statement:** this paper does not encourage token speculation, promises no Token appreciation or returns, and is not investment advice. Credit is a non-transferable internal contribution-accounting unit with no monetary value. Token (if needed) must be compliantly issued, and fiat/stablecoin channels need KYC/AML compliance.
7. **Reproduction requirements:** to reproduce this paper's design proposals, one needs to: ① deploy libp2p nodes and Kademlia DHT; ② implement Agent Manifest content addressing and DHT registration; ③ implement Task Envelope publishing and matching; ④ implement at least L0 sampling and L1 redundant verification; ⑤ implement credit accounting and Reputation aggregation; ⑥ test task success rate, verification-cost ratio, and unit task cost at 100-1000 node scale. Reproduction results should be compared with this paper's key-metric targets.
8. **Limitations:** this paper is a theory-and-design paper and contains no experimental data from real P2P agent networks. All judgments about P2P network performance, cost, and scale are based on literature analysis and theoretical derivation and need verification in real testnets. This paper's critical analyses (like Bittensor's subnet quality problems, verification-cost death sentence) may change with technological progress and need continuous tracking.

This paper is the UDOS v7.7.8 independent supplementary volume (P39), not merged into v7.7.2's 13-paper master volume. Author: Fang Wenxin. Date: 2026-09-21. Twin Earth • UDOS Reasoning Engine.